Supplement: The problem and promise of modeling plant leafout under warmer winters

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The search for better indicators

Increasing evidence suggests that callose and its degradation through 1,3-β-glucanases may control bud endodormancy (Rinne *et al.*, 2011; van der Schoot *et al.*, 2014; Pan *et al.*, 2023; Pandey *et al.*, 2026), potentially alongside phytohormones (such as ABA and GA, Tylewicz *et al.*, 2018; Pan *et al.*, 2021)—but how useful any of these will be as measurable indicators is unclear. Approaches focused on measuring these compounds directly, such as callose deposition and degradation in the bud, or changing concentrations of ABA and GA, are far from simple and unlikely to prove successful alone. One major challenge is that all the compounds found so far are also integral to other plant development and defense pathways, and thus methods need to disentangle these other processes separate from bud dormancy and budbreak.

Using transcriptomic approaches may be one possible way forward. These approaches estimate the relative transcript expression levels of genes implicated in the dormancy cycle, which can be used as proxies for callose, ABA, and GA while concurrently providing molecular insights into the underlying mechanisms responsible. A particular transcript or ratio of transcripts would likely be far easier to measure than callose, ABA, or GA on their own. Rather than attempting to deduce the probability that callose, ABA, or GA are responsible for a particular outcome in a particular phenological pathway, transcriptomics-based approaches can help detect and quantify candidate genes involved in budbreak via temporal and environment-induced changes and alignments in transcript expression that are specific to dormancy and budbreak. These tools can then be combined with others—such as visualizing the gene product to observe subcellular localization, as well as finer-scale methods, such as testing gene function and interaction through experiments with gene mutants.

For example, Zhang *et al.* (2023) used quantitative PCR (one type of transcriptomics), molecular visualization, and ABA treatments to connect several genes to how ABA may control bud dormancy

in peaches (Fig. S1). Using this suite of approaches they showed that a DAM gene called PpDAM6 (MADS-box dormancy associated genes, which are important central regulators of bud dormancy in perennial species Falavigna *et al.*, 2019; Yang *et al.*, 2021) and an associated phosphorylator, called PpMAPK6 (a mitogen-activated protein kinase) are both regulated by ABA. Experiments showed that ABA upregulated PpDAM6 expression and simultaneously inhibited PpMAPK6, sustaining endodormancy. Reduced ABA concentration as endodormancy progressed toward ecodormancy led to the loss of suppression of PpMAPK6, putatively enabling the phosphorylation of PpDAM6 that marks it for degradation and thereby promotes the transition from endodormancy to ecodormancy. Zhang *et al.* (2023) therefore illustrated a combined method of first screening for candidate genes implicated in the dormancy release phenomenon, then tested for their interactions with exogenous ABA application. This clarified the roles of three agents within the dormancy cycle. Thus, at least one of these genes could be a useful marker for endodormancy maintenance given its observed relative transcript levels over time and interaction with ABA, and was also used as a measurable, specific proxy for one of ABA's many roles in this system.

These same concepts apply to callose and GA. For example, 1,3-β-glucanases localized to the plasmodesmata have displayed upregulated expression during the overwintering/endodormancy period. Enzyme levels or activity or, more easily, the transcripts could perhaps be used as a proxy measure of deposition and degradation. This deposition and the closure of plasmodesmata also appear to be mediated by ABA (Tylewicz *et al.*, 2018). Prolonged chilling results in the upregulation of GA biosynthesis, which in turn induces the upregulation of some 1,3-β-glucanases responsible for plasmodesmata reopening (Rinne *et al.*, 2011; Singh *et al.*, 2018). As such, the 1,3--glucanase family could be yet another targetable, clear indicator for modelling dormancy progression and budbreak; members of this family respond to a specific substrate (callose), can be visualized both transcriptomically (e.g., through qPCR) and through reporters to visualize their expression patterns, and are localized to specific tissues.

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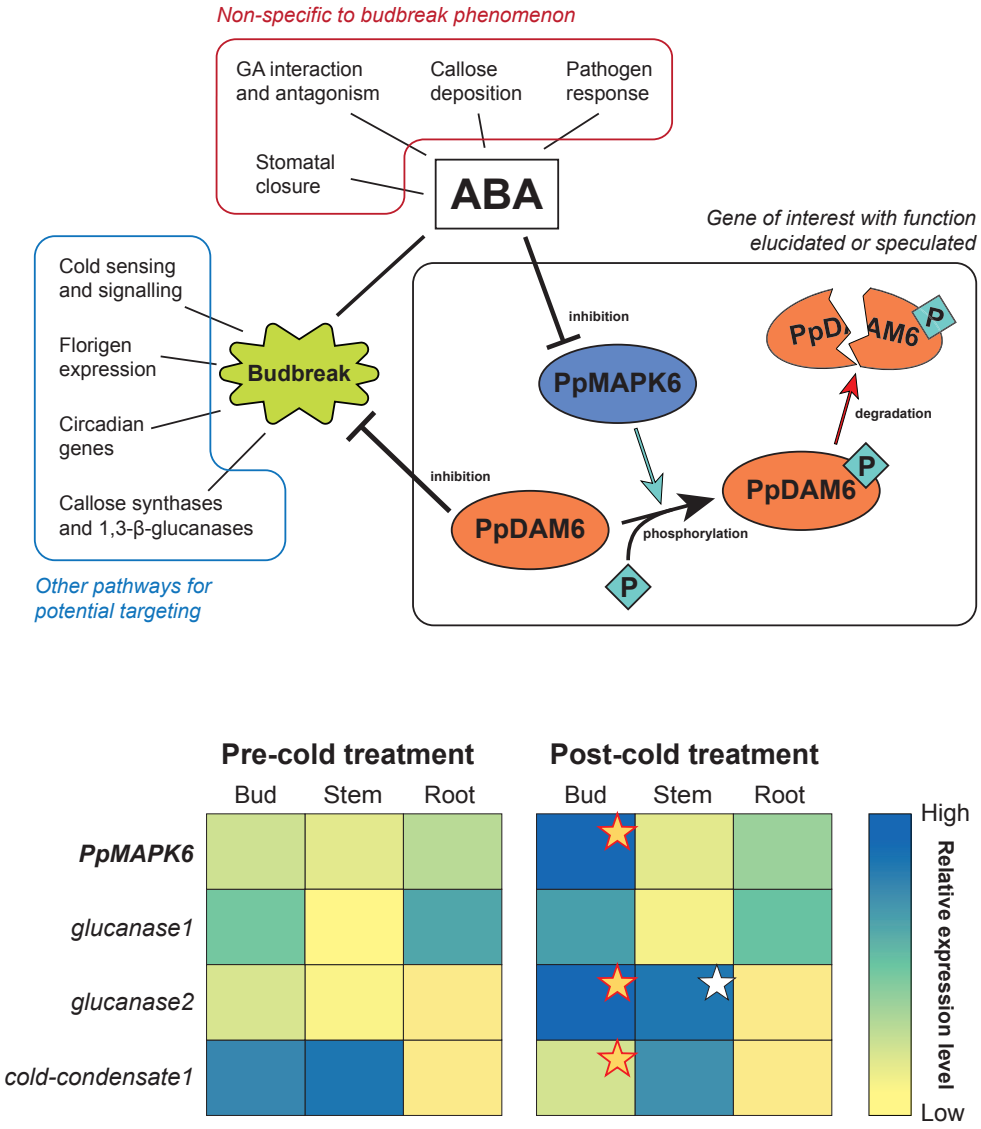


Figure S1: A useful measurable indicator of dormancy, or dormancy break, may come from better understanding the molecular and cellular pathways of budbreak using multiple approaches. Here we show (top) a pathway diagram reflecting what Zhang *et al.* (2023) found alongside a the potential connections of PpDAM6 and PpMAK6 with ABA (noting the other processes related to ABA that are not directly related to budbreak; and (below) the expected qPCR (visualized via a heatmap) of relative transcript levels in different tissues and based on cold ('chilling') treatments.